

8/09/26

classmate

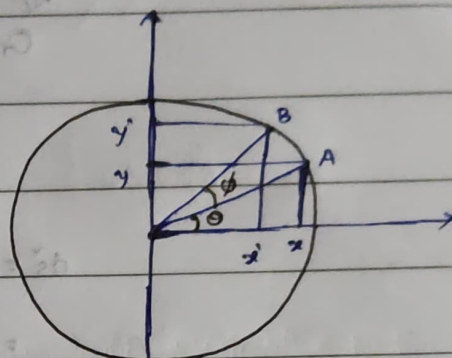
Date

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$$ds^2 = dx^2 + dy^2 + dz^2 - c^2 dt^2$$

$$= c^2 dt^2 - dx^2 - dy^2 - dz^2$$

Invariant under Lorentz Transformation.



Length of radius is invariant.

This is called

Mathematically,

Orthogonal transformation



$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

O

$$\text{Here, } \sqrt{x'^2 + y'^2} = \sqrt{(x')^2 + (y')^2} = r$$

$$O^T O = I$$

$$x'_i = O_{ij} x_j = \sum_j O_{ij} x_j$$

$$\begin{pmatrix} x_1 = x \\ x_2 = y \end{pmatrix}$$

$$x'_1 = O_{11} x_1 + O_{12} x_2$$

$$= \cos\phi x - \sin\phi y$$

Getting into 3D.

$$x_1 = x$$

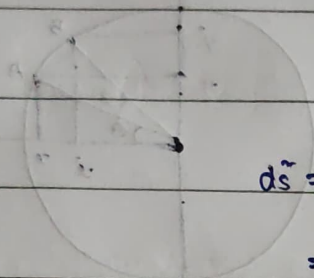
$$x_2 = y$$

$$x_3 = z$$

$$x'_i = \sum_j R_{ij} x_j$$

such that $R^T R = I$ 

Orthogonal transformation.



$$d\tilde{s} = d\tilde{x}^2 + d\tilde{y}^2 + d\tilde{z}^2 - c^2 d\tilde{t}^2$$

$$= dx_1^2 + dy_1^2 + dz_1^2 - c^2 dt_1^2 = ds^2$$

$$(x_1 = x \quad x_2 = y \quad x_3 = z \quad x_0 = ict)$$

$$(\because i = \sqrt{-1})$$

$$= dx_1^2 + dx_2^2 + dx_3^2 + dx_0^2$$

Lorentz vector:

$$A_\mu, \text{ where } \mu = 0, 1, 2, 3$$

spatial component

temporal component (time component)

invariant under

Lorentz Transformation

$$A^2 = A_\mu A^\mu \quad (\text{Lorentz scalar}) \quad (\cong \vec{A} \cdot \vec{A} \text{ in 3D})$$

$$= A_0 A^0 + A_1 A^1 + A_2 A^2 + A_3 A^3$$

$$x_1 = x$$

$$x_2 = y$$

$$x_3 = z$$

$$x_0 = ict$$

Minkowski space

Prescription:
$$\left. \begin{aligned} A_i &\rightarrow A^i \\ A_0 &\rightarrow -A^0 \end{aligned} \right\} \begin{aligned} A_i &\rightarrow -A^i \\ A_0 &\rightarrow A^0 \end{aligned}$$

$$\tilde{x} = x^\mu x_\mu$$

$$= x^0 x_0 + x^1 x_1 + x^2 x_2 + x^3 x_3$$

$$= -x_0^2 + x_1^2 + x_2^2 + x_3^2$$

$$= x_0^2 - x_1^2 - x_2^2 - x_3^2$$

A_μ, B_μ } Lorentz vectors $\Rightarrow A_\mu A^\mu, B_\mu B^\mu, A^\mu B_\mu$ } Lorentz scalars

Generalised Lorentz Transformation

$$\vec{x} = \underbrace{x_\parallel}_{\text{parallel to } \vec{v}} + \underbrace{x_\perp}_{\text{perpendicular to } \vec{v}}$$

Exercise: Find the transformation matrix in this case.

$$\vec{x}' = x'_\parallel + x'_\perp$$

$$x'_\perp = x_\perp$$

$$x'_\parallel = \gamma(x_\parallel - \vec{v}t)$$

$$\vec{x}' = \gamma \vec{x}_\parallel - \gamma \vec{v}t + \vec{x}_\perp$$